

Safety Procedure & Emergency Response Manual

Mechanical Maintenance Work - Coastal Ridge Wind Farm - Revision 6

1. Purpose & Scope

This procedure defines the minimum safety requirements for all personnel performing mechanical or electrical maintenance work at Coastal Ridge Wind Farm, including turbine climbing, nacelle work, ground-level equipment servicing, and substation access. It applies to site employees, contractors, and OEM technicians alike. Anyone who has not completed the site safety induction may not perform maintenance work unsupervised.

2. Personal Protective Equipment Requirements

All personnel must wear the following personal protective equipment while on the turbine pad or inside any turbine, substation, or workshop: a certified hard hat, safety glasses or a full face shield when using rotating tools, cut-resistant gloves rated to the task, steel-toe safety boots, and hearing protection inside the nacelle or near running equipment. Anyone climbing above ground level, including nacelle and hub access, must additionally wear a certified full-body fall-arrest harness connected to an approved anchor point at all times above the first platform.

Damaged or expired PPE must be removed from service immediately and reported to the Site Safety Officer. Harnesses involved in any fall-arrest event, however minor, must be retired and replaced, not reused, regardless of visible condition.

3. Lockout-Tagout (LOTO) Procedure

- Identify all energy sources for the equipment being serviced (electrical, hydraulic, mechanical stored energy in the rotor and yaw system).
- Notify affected personnel before isolating equipment.
- Apply locks and tags to each isolation point; each technician working on the equipment applies their own personal lock.
- Verify zero energy state before beginning work, including a manual attempt to operate the equipment's normal controls.
- Remove locks and tags only after work is complete and all tools and personnel are clear of the equipment.

4. Working at Height Requirements

Before every climb, technicians must visually inspect their harness, lanyard, and connectors for cuts, fraying, or corrosion, and confirm the harness has not exceeded its service life or been involved in a prior fall-arrest event. Technicians must maintain 100% tie-off at all times above the first internal platform, meaning a second point of connection must be established before disconnecting the first when transitioning between anchor points.

Fall clearance must be verified before working near an open hatch or edge: a minimum of 6 metres of clear fall path is required beneath any working position where a fall-arrest system would need to deploy. Rescue equipment must be available on-site and at least one person on the climbing team must hold a current suspension-trauma rescue certification.

5. Hazard Identification & Risk Assessment

A task-specific risk assessment must be completed and signed before any non-routine maintenance task begins. Routine, pre-assessed tasks (such as the standard bi-weekly inspection rotation) are covered by the site's standing risk assessment, which is reviewed quarterly or immediately after any near-miss or incident.

6. Confined Space Entry

The turbine tower base cable vault and the substation cable trench are both classified confined spaces. Entry requires a confined space permit, atmospheric testing before entry, continuous ventilation, and a dedicated attendant stationed at the entry point for the full duration of the work, in addition to standard PPE.

7. Incident & Near-Miss Reporting

Any incident, injury, or near-miss must be reported to the Site Safety Officer immediately, and in all cases within the same shift. A near-miss is any event that could have resulted in injury or damage under slightly different circumstances, even if no harm occurred - for example, a piece of equipment found not to specification after a technician had already relied on it. Under-reporting near-misses is treated as a safety violation in itself, since these reports are the primary way the site catches systemic issues before they cause an injury.

8. Emergency Response

8.1 Medical Emergency

Call the site emergency line immediately, then notify the Site Safety Officer and the on-duty Supervisor. Do not move an injured person unless there is an immediate life-threatening hazard. If the injury occurs at height, initiate the suspension-trauma rescue procedure without delay - suspension trauma can become life-threatening within minutes.

8.2 Fire

Activate the nearest fire alarm and evacuate to the designated muster point. Do not attempt to fight an electrical or nacelle fire unless trained and equipped to do so; nacelle fires are generally not survivable to fight directly given the height and confined space.

8.3 Severe Weather

All climbing work must stop and personnel must descend immediately when wind speed exceeds 18 m/s at hub height, or when lightning is detected within 10 km of the site.

9. Safety Bulletin - Mandatory Update (Effective Immediately)

CRITICAL - Following a near-miss on September 9, 2026 at Turbine T-22, in which a technician briefly clipped into an internal fall-arrest anchor point before discovering that one of its mounting bolts was loose, the Q3 Operations Review approved an immediate revision to the pre-climb inspection requirement site-wide.

Effective immediately, every technician must perform a physical torque-check - not only a visual check - of both the primary and backup internal anchor points before clipping in for any climb, using the calibrated torque wrench kept at each turbine base cabinet. A visual-only inspection is no longer sufficient and does not satisfy this procedure.

All climbing personnel, including contractors and OEM technicians, must complete a refresher briefing on this updated procedure before their next climb. The Site Safety Officer will track completion and has authority to suspend climb authorization for anyone who has not completed the briefing. A full anchor-point re-certification audit of all 36 turbines must be completed by September 25, 2026. Owner: Site Safety Officer (D. Petrov). Priority: Critical.

10. Roles & Responsibilities

Role	Responsibility
Site Safety Officer	Owns this procedure, investigates incidents and near-misses, authorises or suspends climb access
Maintenance Lead	Ensures crews follow LOTO and height-work requirements on every task
Technician	Performs pre-task PPE and equipment inspection; reports hazards and near-misses immediately
Site Operations Manager	Approves procedure revisions and resourcing for corrective actions

11. Revision History

Revision	Date	Summary of change
4	2025-11-02	Added confined space entry section
5	2026-03-14	Updated severe weather thresholds
6	2026-09-15	Added mandatory anchor-point torque-check bulletin following T-22 near-miss